

Self-Learning: Address Decoding Techniques & 8259 Cascading — MPCA Module 4

Definition: Address Decoding Techniques define how system address lines are mapped to peripheral chip selects (Absolute vs Partial), while **8259 Cascading** expands interrupt handling capacity from 8 to 64 priority channels.

1. Absolute vs Partial Address Decoding

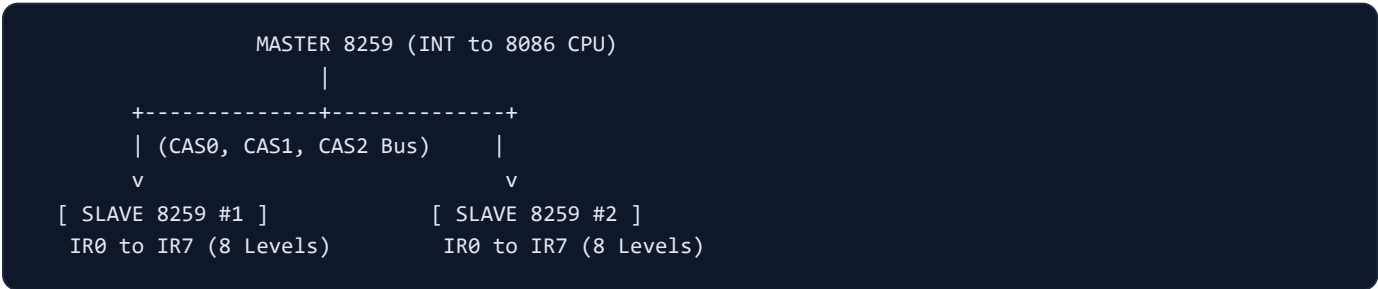
- ABSOLUTE ADDRESS DECODING:
- ALL address lines (A0 - A19) are fully connected to decoders and chip logic.
 - Each memory/IO location has exactly ONE unique physical address.
 - Eliminates "Shadow Addresses" (ghost aliases).
- PARTIAL ADDRESS DECODING:
- Only some upper address lines are decoded; unused "don't care" lines (X) are left unconnected.
 - Simpler and cheaper hardware (fewer logic gates).
 - Causes "Shadow Addresses" (same physical hardware responds to multiple address ranges).

Comparison Table:

Feature	Absolute Decoding	Partial Decoding
Address Line Usage	All address lines (A_0 - A_19) fully decoded.	Only subset of lines decoded (A_15-A_19).
Shadow Addresses	Zero Shadow Addresses (Strict 1-to-1 mapping).	Multiple shadow aliases exist.
Hardware Complexity	High (requires multi-input logic gates/decoders).	Low (simple 74LS138 decoder).
Best Suited For	Large systems, 1 MB full memory maps.	Small microcontroller/embedded systems.

2. 8259 Cascaded Mode (Master-Slave Configuration)

To handle more than 8 interrupts, one **Master 8259** is connected to up to **8 Slave 8259s** via the 3-bit Cascade Bus (**CAS_0, CAS_1, CAS_2**).



Cascading Operation Flow:

1. Slave 8259 receives interrupt on its IR line and signals Master 8259 via Master's IR pin.
 2. Master sends `INT` to 8086 CPU.
 3. 8086 issues 1st INTA. Master broadcasts the 3-bit Slave ID on **CAS_0 - CAS_2**.
 4. The matching Slave recognizes its ID, and on the 2nd INTA, the **Slave 8259 puts the interrupt vector byte** onto the Data Bus.
 5. Max Capacity = 1 Master × 8 Slaves = **64 Interrupt Inputs**.
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3. Quick Recall Flow

Absolute (All Lines, No Shadows) vs Partial (Don't Cares, Shadow Addresses) | 8259 Cascading: Master-Slave via CAS0-2 -> Up to 64 Interrupts